

Applications (Part II) Homework

Name
Date
Course

① The population of a colony of bacteria follows the law of uninhibited growth ($k=0.18$). If a colony starts with 103 bacteria, how many will be present after 9 days?

② Sodium-24 converts to Magnesium according to law of uninhibited decay ($k=-0.0462$). If 3.22 grams of sodium-24 is present in a sample, how much would be left after 7.2 hours?

③ The population of a culture of cells grows from 157 to 366 in 42 minutes. If this same strain of cells starts with 400 cells, how many will there be after 235 minutes?

④ In 4 years, a radioactive substance decreases from 100 grams to 74 grams. How much of this same radioactive material would be left if we started with a 71 gram sample and monitored it for 2 years?

⑤ The half life of Krypton-94 is 1.4 seconds. If a scientist obtains 2.7 grams of Krypton-94, how much will be left after 3.8 seconds?

⑥ The half-life of hydrogen-3 is 12.3 years. If a physicist obtains 0.98 grams, how much will be left after 51 years?

⑦ The number of organisms in a suspension of yeast doubles in 83 minutes. How long will it take for the suspension to grow from 100 organisms to 379 organisms?

⑧ After 14.4 seconds, a radioactive material retains 84% of its original mass. What is the half life of this radioactive material?

⑨ The population of a suspension of yeast after t hours can be predicted with the equation below.

$$P(t) = \frac{2,000}{1 + 31e^{-0.25t}}$$

a) What is the carrying capacity of the population?

b) What is the initial population?

c) What will the population be after 8 hours?

⑩ The population of a colony of bacteria after t minutes can be predicted with the equation below.

$$P(t) = \frac{23,455}{1 + 20e^{-0.27t}}$$

a) What is the carrying capacity of the population?

b) What is the initial population?

c) What will the population be after 141 minutes?

Homework Answers

- ① 520 bacteria
- ② 2.3 grams sodium-24
- ③ 45,579 cells
- ④ 61 grams
- ⑤ 0.41 grams krypton-94
- ⑥ 0.0553 grams hydrogen-3
- ⑦ 159.5 minutes
- ⑧ 57.2 seconds
- ⑨ a) 2,000 organisms
b) 62.5 organisms
c) 385 organisms
- ⑩ a) 23,455 bacteria
b) 1,117 bacteria
c) 23,455 bacteria